ACADEMIC PROJECT REPORT

CORE BANKING MANAGEMENT SYSTEM

Implementation using Binary Search Tree, Stack, and Circular Queue in C++

Subject: Data Structures (Course Code: `102040304`)

Department: Department of Computer Engineering / Computer Science & Design Engineering

Institution: G H Patel College of Engineering & Technology (GCET), CVM University

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1. Introduction & Objectives

1.1 Project Overview

In modern retail banking, customer satisfaction and operational speed rely on efficient management of customer accounts, transaction auditing, and in-branch visitor traffic.

The **Core Banking Management System** is a lightweight, high-performance console application built to simulate core banking operations while demonstrating the practical application of fundamental data structures covered in the semester syllabus.

1.2 Key Objectives

1. **Fast Account Indexing & Retrieval:** Maintain dynamic customer records with $O(\log n)$ average search complexity using a **Binary Search Tree (BST)**.

2. **Real-Time Transaction Logging:** Keep an active audit trail of recent deposits and withdrawals using the **Last-In, First-Out (LIFO)** property of a **Stack**.

3. **Fair In-Branch Customer Traffic Control:** Ensure orderly visitor service using the **First-In, First-Out (FIFO)** property of a **Circular Queue**.

4. **Code Elegance & Maintainability:** Deliver the entire modular architecture within ~150 lines of clear, standard C++ without external template libraries (STL), demonstrating raw algorithmic implementation.

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2. System Architecture & Data Structures Mapping

The project integrates three distinct data structures, each chosen to address a specific real-world requirement:

```
                          CORE BANKING SYSTEM
                                  |
    +-----------------------------+-------------------------------+
    |                             |                               |
[Accounts Module]          [Transaction History]          [Token Waiting Line]
    |                             |                               |
Binary Search Tree (BST)       Stack (Array)               Circular Queue (Array)
    |                             |                               |
Fast Search: O(log n)          LIFO Principle                  FIFO Principle
Sorted Output: Inorder         Most Recent First               Reuses Memory (% MA
X)
```

2.1 Binary Search Tree (BST) — *Practical 8*

Application: Storage and indexing of bank accounts keyed by `accNo`.

Why BST?

* In a linear array or linked list, searching for an account requires checking every node sequentially ($O(n)$ time).

* In a BST, each comparison cuts the search space in half, achieving an average lookup time of $O(\log n)$.

* **Inorder Traversal (Left $\to$ Root $\to$ Right)** guarantees that customer records are retrieved in strictly sorted numerical order of Account Number without requiring an external sorting algorithm.

2.2 Stack — *Practical 3*

Application: ATM Mini-Statement generation.

Why Stack?

* When a customer requests a mini-statement from an ATM or mobile app, they expect to see their **most recent** transaction first.

* This matches the **Last-In, First-Out (LIFO)** behavior of a Stack. New transactions are pushed to the top, and printing pops/displays from top to bottom.

2.3 Circular Queue — *Practical 5B*

Application: Branch visitor token queue and teller desk allocation.

Why Circular Queue?

* Service desks must operate under strict **First-In, First-Out (FIFO)** fairness: the customer who arrived first must be served first.

* In a traditional linear queue implemented with an array, dequeuing elements leaves empty spaces at the front that cannot be reused, causing a false "Queue Full" condition.

* The circular queue uses modulo arithmetic `(rear + 1) % MAX` and `(front + 1) % MAX` to wrap pointers around, continuously reusing array memory.

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3. Function-by-Function Description

3.1 Account Management Functions (BST)

1. `Account* createAccount(int accNo, string name, float bal)`

Purpose: Dynamic memory allocation for a new customer record.

How it Works: Uses the `new` operator to allocate a node of type `struct Account`, initializes account number, name, and opening balance, and sets child pointers (`left` and `right`) to `NULL`.

2. `Account *insert(Account* root, int accNo, string name, float bal)`

Purpose: Inserts an account node at its correct position in the BST.

How it Works: Compares `accNo` with the current node:

* If `accNo < root->accNo`, recurses to the **left** subtree.

* If `accNo > root->accNo`, recurses to the **right** subtree.

* If root is `NULL`, creates and returns the new node.

3. `Account *search(Account* root, int accNo)`

Purpose: Locates an existing customer account by ID.

How it Works: Traverses downward from the root. If `accNo` matches, it returns the pointer to that node. If smaller, it searches left; if larger, it searches right. Returns `NULL` if not found.

4. `void inorder(Account* root)`

Purpose: Displays all bank accounts in ascending order.

How it Works: Recursively visits:

1. Left subtree (`inorder(root->left)`)

2. Root node (prints account details)

3. Right subtree (`inorder(root->right)`)

3.2 Transaction History Functions (Stack)

5. `void pushTrans(float amt)`

Purpose: Records a financial event onto the transaction stack.

How it Works:

* Positive `amt` represents a **Deposit**; negative `amt` represents a **Withdrawal**.

* If the stack reaches `MAX - 1`, older entries are shifted up by 1 position so the latest 5 transactions are preserved.

* Increments `top` and stores the amount at `transStack[top]`.

6. `void printMiniStatement()`

Purpose: Outputs the mini-statement in LIFO order.

How it Works: Loops backwards from `i = top` down to `0`. If `transStack[i] > 0`, it prints `Deposited: Rs. X`; otherwise, it prints `Withdrawn: Rs. X`.

3.3 Customer Service Desk Functions (Circular Queue)

7. `void issueToken(int token)`

Purpose: Enqueues a customer into the branch waiting line.

How it Works: Checks for overflow condition `(rear + 1) % MAX == front`. If not full, increments `rear = (rear + 1) % MAX`, writes `token` at `tokenQueue[rear]`, and sets `front = 0` if it was previously empty.

8. `void serveCustomer()`

Purpose: Dequeues the next customer to be served by the teller.

How it Works: Checks for underflow (`front == -1`). If customers are present, prints `Calling Token #tokenQueue[front]`. If `front == rear`, resets both to `-1` (queue now empty); otherwise advances `front = (front + 1) % MAX`.

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4. Real-World Practicality & Benefits

1. **Scalability:** The BST structure allows a bank to handle hundreds of thousands of accounts with logarithmic lookup times without linear scan bottlenecks.

2. **Resource Efficiency:** The circular queue guarantees zero memory leakage and zero wasted array slots in fixed-memory hardware devices (such as branch kiosk ticket printers).

3. **Data Integrity:** Strict encapsulation ensures balances cannot be modified without being logged into the transaction stack.

4. **Standardization:** Avoids third-party dependencies, making it directly portable across embedded devices, legacy ATMs, and modern operating systems.

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5. Complete C++ Source Code (`bank_system.cpp`)

```
#include <iostream>
#include <string>

using namespace std;

#define MAX 5

// --- 1. BINARY SEARCH TREE (BST) - Account Records (Practical 8) ---
struct Account {
    int accNo;
    string name;
    float balance;
    Account left, right;
};

// Create a new account node
Account* createAccount(int accNo, string name, float bal) {
    Account* n = new Account();
    n->accNo = accNo;
    n->name = name;
    n->balance = bal;
    n->left = n->right = NULL;
    return n;
}

// Insert account into BST (ordered by accNo)
Account insert(Account root, int accNo, string name, float bal) {
    if (!root) return createAccount(accNo, name, bal);
```

```
    if (accNo < root->accNo) root->left = insert(root->left, accNo, name, bal);
    else if (accNo > root->accNo) root->right = insert(root->right, accNo, name, ba
l);
    return root;
}

// Search account by accNo in O(log n)
Account search(Account root, int accNo) {
    if (!root || root->accNo == accNo) return root;
    if (accNo < root->accNo) return search(root->left, accNo);
    return search(root->right, accNo);
}

// Inorder Traversal: Left -> Root -> Right (prints sorted accounts)
void inorder(Account* root) {
    if (root) {
        inorder(root->left);
        cout << "A/C: " << root->accNo << " | Name: " << root->name << " | Balance:
Rs. " << root->balance << endl;
        inorder(root->right);
    }
}

// --- 2. STACK - ATM Mini-Statement (Practical 3: LIFO) ---
float transStack[MAX];
int top = -1;

void pushTrans(float amt) {
    if (top == MAX - 1) { // Shift up if full to keep last 5 transactions
        for (int i = 0; i < MAX - 1; i++) transStack[i] = transStack[i + 1];
        top--;
    }
    transStack[++top] = amt;
}

void printMiniStatement() {
    if (top == -1) { cout << "\nNo transactions recorded yet!" << endl; return; }
    cout << "\n--- ATM MINI-STATEMENT (LIFO: Most Recent First) ---" << endl;
    for (int i = top; i >= 0; i--) {
        if (transStack[i] > 0)
            cout << "[" << (top - i + 1) << "] Deposited: Rs. " << transStack[i] <<
endl;
        else
            cout << "[" << (top - i + 1) << "] Withdrawn: Rs. " << -transStack[i] <
< endl;
    }
}

// --- 3. CIRCULAR QUEUE - Customer Token Line (Practical 5B: FIFO) ---
int tokenQueue[MAX];
```

```
int front = -1, rear = -1;

void issueToken(int token) {
    if ((rear + 1) % MAX == front) { cout << "Queue is full! Wait for counter to be
free." << endl; return; }
    if (front == -1) front = 0;
    rear = (rear + 1) % MAX;
    tokenQueue[rear] = token;
    cout << "Token #" << token << " issued! Please wait in customer lounge." << end
l;
}

void serveCustomer() {
    if (front == -1) { cout << "No customers waiting in queue!" << endl; return; }
    cout << ">> Calling Token #" << tokenQueue[front] << " to Teller Counter 1 (FIF
O)" << endl;
    if (front == rear) front = rear = -1;
    else front = (front + 1) % MAX;
}

// --- MAIN MENU ---
int main() {
    Account* root = NULL;
    int choice, accNo, nextToken = 101;
    string name;
    float amount;

    // Pre-loaded sample accounts for quick viva demo
    root = insert(root, 102, "Suresh", 15000.0f);
    root = insert(root, 101, "Ramesh", 8500.0f);
    root = insert(root, 105, "Priya", 24000.0f);
    root = insert(root, 103, "Anjali", 12000.0f);

    while (true) {
        cout << "\n=== CORE BANKING SYSTEM (DS PROJECT - C++) ===" << endl;
        cout << "1. Open Account (BST Insert)     2. Search Account (BST Search)" <
< endl;
        cout << "3. Display All (BST Inorder)     4. Deposit Money (Stack Record)"
<< endl;
        cout << "5. Withdraw Money (Stack Record) 6. Mini-Statement (Stack LIFO)" <
< endl;
        cout << "7. Issue Token (Queue Enqueue)   8. Serve Customer (Queue Dequeu
e)" << endl;
        cout << "9. Exit" << endl;
        cout << "Choice (1-9): ";
        if (!(cin >> choice)) break;

        if (choice == 1) {
            cout << "Enter A/C No, Name, Initial Deposit: ";
            cin >> accNo >> name >> amount;
```

```
            root = insert(root, accNo, name, amount);
            cout << "Account " << accNo << " created successfully!" << endl;
        } else if (choice == 2) {
            cout << "Enter A/C No to Search: "; cin >> accNo;
            Account* a = search(root, accNo);
            if (a) cout << "Found! Name: " << a->name << " | Balance: Rs. " << a->b
alance << endl;
            else cout << "Account not found!" << endl;
        } else if (choice == 3) {
            cout << "\n--- All Bank Accounts (Sorted by A/C No) ---" << endl;
            inorder(root);
        } else if (choice == 4 || choice == 5) {
            cout << "Enter A/C No and Amount: "; cin >> accNo >> amount;
            Account* a = search(root, accNo);
            if (!a) { cout << "Account not found!" << endl; continue; }

            if (choice == 4) {
                a->balance += amount;
                pushTrans(amount);
                cout << "Deposit successful! New Balance: Rs. " << a->balance << en
dl;
            } else {
                if (a->balance < amount) cout << "Insufficient balance! Current: R
s. " << a->balance << endl;
                else {
                    a->balance -= amount;
                    pushTrans(-amount);
                    cout << "Withdrawal successful! New Balance: Rs. " << a->balanc
e << endl;
                }
            }
        } else if (choice == 6) {
            printMiniStatement();
        } else if (choice == 7) {
            issueToken(nextToken++);
        } else if (choice == 8) {
            serveCustomer();
        } else if (choice == 9) {
            cout << "Thank you for using the Banking System!" << endl;
            break;
        } else {
            cout << "Invalid choice! Please select 1-9." << endl;
        }
    }
    return 0;
}
```

6. Detailed Step-by-Step Test Output

Below is the verified console session demonstrating each feature:

```
=== CORE BANKING SYSTEM (DS PROJECT - C++) ===
1. Open Account (BST Insert)     2. Search Account (BST Search)
3. Display All (BST Inorder)     4. Deposit Money (Stack Record)
5. Withdraw Money (Stack Record) 6. Mini-Statement (Stack LIFO)
7. Issue Token (Queue Enqueue)   8. Serve Customer (Queue Dequeue)
9. Exit
Choice (1-9): 3

--- All Bank Accounts (Sorted by A/C No) ---
A/C: 101 | Name: Ramesh | Balance: Rs. 8500
A/C: 102 | Name: Suresh | Balance: Rs. 15000
A/C: 103 | Name: Anjali | Balance: Rs. 12000
A/C: 105 | Name: Priya | Balance: Rs. 24000

=== CORE BANKING SYSTEM (DS PROJECT - C++) ===
Choice (1-9): 2
Enter A/C No to Search: 103
Found! Name: Anjali | Balance: Rs. 12000

=== CORE BANKING SYSTEM (DS PROJECT - C++) ===
Choice (1-9): 4
Enter A/C No and Amount: 101 1500
Deposit successful! New Balance: Rs. 10000

=== CORE BANKING SYSTEM (DS PROJECT - C++) ===
Choice (1-9): 5
Enter A/C No and Amount: 101 2000
Withdrawal successful! New Balance: Rs. 8000

=== CORE BANKING SYSTEM (DS PROJECT - C++) ===
Choice (1-9): 6

--- ATM MINI-STATEMENT (LIFO: Most Recent First) ---
[1] Withdrawn: Rs. 2000
[2] Deposited: Rs. 1500

=== CORE BANKING SYSTEM (DS PROJECT - C++) ===
Choice (1-9): 7
Token #101 issued! Please wait in customer lounge.

=== CORE BANKING SYSTEM (DS PROJECT - C++) ===
Choice (1-9): 7
Token #102 issued! Please wait in customer lounge.
```

```
=== CORE BANKING SYSTEM (DS PROJECT - C++) ===
Choice (1-9): 8
>> Calling Token #101 to Teller Counter 1 (FIFO)

=== CORE BANKING SYSTEM (DS PROJECT - C++) ===
Choice (1-9): 8
>> Calling Token #102 to Teller Counter 1 (FIFO)

=== CORE BANKING SYSTEM (DS PROJECT - C++) ===
Choice (1-9): 9
Thank you for using the Banking System!
```

---

7. Viva Preparation Guide & Common Questions

1. **Why is BST better than a simple array for bank accounts?**

Answer:* Searching in an array takes $O(n)$ because we scan sequentially. In a BST, searching takes $O(\log n)$ because every comparison divides the search space into half. Furthermore, an **Inorder Traversal** prints the accounts in sorted order automatically without extra sorting steps.

2. **Why does the ATM Mini-Statement use a Stack?**

Answer:* A bank mini-statement must present the most recent transaction at the top. The Stack follows **Last-In, First-Out (LIFO)**, meaning the last pushed transaction is the first one displayed.

3. **What is the mathematical condition for Circular Queue full?**

Answer:* `(rear + 1) % MAX == front`. This detects when the rear pointer has wrapped around and reached the position right behind front.

4. **What happens if a customer attempts to withdraw more than their balance?**

Answer:* The system executes an account balance check: `if (a->balance < amount)`. It blocks the withdrawal, preserves the current balance, and suppresses recording the transaction in the stack.

Report prepared for Semester 3 Data Structures Mini-Project Submission.